

## BRIGNOGAN, France

WMO: 071070

Lat: 48.676N

Lon: 4.331W

Elev: 18

StdP: 101.11

Time Zone: 1.00 (EUC)

Period: 99-18

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 1.3        | 2.7 | -3.1                          | 2.9 | 3.0  | -1.6 | 3.3 | 4.0  | 20.8                  | 7.6  | 19.2 | 7.7  | 4.9                   | 110  | 0.688 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 5.0                    | 21.7            | 17.5 | 20.4 | 17.1 | 19.6 | 16.8 | 18.5                | 20.5 | 17.8 | 19.6 | 17.3 | 19.0 | 6.0                  | 190  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 17.7                            | 12.7 | 19.3 | 17.1 | 12.2 | 18.8 | 16.5 | 11.8 | 18.3 | 52.3          | 20.6 | 50.3 | 19.7 | 48.7 | 19.1 | 21.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 17.2              | 14.8 | 13.1 | -0.8                       | 26.8 | 1.2                | 2.5 | -1.7                                               | 28.5 | -2.4       | 30.0 | -3.1       | 31.4 | -4.0       | 33.2 |  |  |
|                   |      |      | -1.9                       | 20.1 | 1.3                | 0.8 | -2.8                                               | 20.7 | -3.6       | 21.2 | -4.3       | 21.6 | -5.2       | 22.2 |  |  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 12.3   | 8.3   | 8.1   | 9.2   | 10.6  | 12.9  | 15.1  | 16.6  | 16.8  | 15.9  | 13.9  | 11.1  | 9.0  |
|                                                                        | DBStd       | 3.76   | 2.52  | 2.38  | 2.26  | 1.83  | 1.64  | 1.57  | 1.38  | 1.33  | 1.74  | 2.16  | 2.10  | 2.66 |
|                                                                        | HDD10.0     | 241    | 61    | 60    | 42    | 13    | 1     | 0     | 0     | 0     | 0     | 1     | 12    | 51   |
|                                                                        | HDD18.3     | 2209   | 309   | 287   | 284   | 232   | 168   | 98    | 56    | 51    | 77    | 140   | 218   | 291  |
|                                                                        | CDD10.0     | 1086   | 10    | 7     | 16    | 31    | 91    | 154   | 206   | 212   | 177   | 121   | 44    | 18   |
|                                                                        | CDD18.3     | 12     | 0     | 0     | 0     | 0     | 0     | 1     | 3     | 4     | 3     | 1     | 0     | 0    |
|                                                                        | CDH23.3     | 24     | 0     | 0     | 0     | 0     | 1     | 5     | 2     | 8     | 7     | 2     | 0     | 0    |
|                                                                        | CDH26.7     | 4      | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 1     | 1     | 1     | 0     | 0    |
| (14) Wind                                                              | WSAvg       | 6.3    | 7.5   | 7.6   | 7.1   | 6.5   | 6.1   | 5.3   | 5.0   | 4.8   | 5.3   | 5.7   | 7.2   | 7.4  |
| (15) Precipitation                                                     | PrecAvg     | 1013   | 122   | 102   | 64    | 76    | 62    | 54    | 60    | 59    | 52    | 108   | 126   | 123  |
|                                                                        | PrecMax     | 1223   | 223   | 212   | 176   | 173   | 125   | 131   | 129   | 144   | 129   | 170   | 244   | 233  |
|                                                                        | PrecMin     | 799    | 29    | 15    | 10    | 16    | 12    | 7     | 11    | 12    | 8     | 22    | 28    | 21   |
|                                                                        | PrecStd     | 126    | 54    | 49    | 41    | 46    | 34    | 33    | 33    | 37    | 30    | 36    | 53    | 59   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 13.9  | 14.2  | 17.1  | 18.2  | 20.6  | 21.8  | 23.3  | 24.2  | 23.3  | 21.4  | 17.1  | 15.3 |
|                                                                        |             | MCWB   | 12.3  | 12.0  | 12.2  | 12.8  | 15.4  | 17.3  | 17.8  | 18.9  | 18.3  | 17.5  | 15.0  | 13.0 |
|                                                                        | 2%          | DB     | 13.2  | 13.0  | 14.2  | 15.4  | 17.7  | 19.6  | 21.0  | 21.2  | 21.0  | 19.0  | 15.8  | 14.2 |
|                                                                        |             | MCWB   | 12.1  | 11.4  | 11.6  | 12.1  | 14.3  | 16.3  | 17.4  | 17.5  | 17.4  | 16.2  | 14.0  | 12.8 |
|                                                                        | 5%          | DB     | 12.5  | 12.2  | 13.1  | 14.3  | 16.3  | 18.6  | 20.0  | 20.0  | 19.5  | 17.8  | 14.8  | 13.4 |
|                                                                        |             | MCWB   | 11.4  | 10.8  | 11.3  | 11.7  | 13.8  | 15.8  | 17.1  | 17.1  | 16.9  | 15.7  | 13.2  | 12.2 |
|                                                                        | 10%         | DB     | 11.8  | 11.5  | 12.2  | 13.3  | 15.4  | 17.7  | 19.1  | 19.2  | 18.6  | 16.9  | 14.1  | 12.6 |
|                                                                        |             | MCWB   | 10.6  | 10.2  | 10.6  | 11.3  | 13.2  | 15.2  | 16.7  | 16.8  | 16.4  | 15.2  | 12.6  | 11.4 |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 13.0  | 12.7  | 13.1  | 13.8  | 16.1  | 18.0  | 19.0  | 19.6  | 19.5  | 17.9  | 15.8  | 13.7 |
|                                                                        |             | MCDB   | 13.4  | 13.6  | 15.3  | 16.5  | 19.2  | 20.6  | 21.2  | 22.4  | 22.1  | 20.4  | 16.8  | 14.3 |
|                                                                        | 2%          | WB     | 12.4  | 11.8  | 12.3  | 12.8  | 14.8  | 16.9  | 18.1  | 18.5  | 18.2  | 16.9  | 14.3  | 13.0 |
|                                                                        |             | MCDB   | 12.9  | 12.5  | 13.7  | 14.6  | 16.8  | 19.0  | 20.1  | 20.1  | 20.1  | 18.4  | 15.3  | 13.9 |
|                                                                        | 5%          | WB     | 11.6  | 11.1  | 11.6  | 12.2  | 14.1  | 16.2  | 17.5  | 17.8  | 17.4  | 16.1  | 13.6  | 12.3 |
|                                                                        |             | MCDB   | 12.3  | 11.9  | 12.9  | 13.8  | 15.8  | 18.0  | 19.4  | 19.3  | 19.1  | 17.5  | 14.6  | 13.3 |
|                                                                        | 10%         | WB     | 11.0  | 10.4  | 10.9  | 11.6  | 13.5  | 15.6  | 17.0  | 17.2  | 16.7  | 15.4  | 12.8  | 11.6 |
|                                                                        |             | MCDB   | 11.7  | 11.4  | 12.1  | 13.1  | 15.1  | 17.2  | 18.7  | 18.8  | 18.2  | 16.6  | 13.9  | 12.6 |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 4.0   | 4.1   | 4.4   | 4.7   | 4.5   | 4.7   | 4.8   | 5.0   | 5.2   | 4.9   | 4.1   | 4.0  |
|                                                                        |             | MCDBR  | 4.0   | 4.6   | 5.8   | 6.4   | 6.2   | 6.2   | 6.1   | 6.2   | 6.9   | 6.0   | 4.9   | 3.9  |
|                                                                        |             | MCWBR  | 4.1   | 3.9   | 3.9   | 3.9   | 3.7   | 3.7   | 3.4   | 3.6   | 4.1   | 4.0   | 4.1   | 3.9  |
|                                                                        | 5% WB       | MCDBR  | 4.0   | 4.2   | 5.2   | 5.6   | 5.5   | 5.7   | 5.5   | 5.5   | 6.1   | 5.4   | 4.6   | 3.9  |
|                                                                        |             | MCWBR  | 4.3   | 4.1   | 3.8   | 3.9   | 3.6   | 3.6   | 3.5   | 3.6   | 3.9   | 4.0   | 4.3   | 4.1  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | 0.339  | 0.353 | 0.380 | 0.387 | 0.391 | 0.383 | 0.385 | 0.373 | 0.364 | 0.360 | 0.340 | 0.332 |      |
|                                                                        | taud        | 2.385  | 2.371 | 2.314 | 2.317 | 2.353 | 2.402 | 2.403 | 2.455 | 2.461 | 2.453 | 2.419 | 2.352 |      |
|                                                                        | Ebn at Noon | 712    | 791   | 828   | 861   | 870   | 878   | 871   | 867   | 839   | 776   | 706   | 661   |      |
|                                                                        | Edh at Noon | 69     | 87    | 107   | 118   | 118   | 114   | 112   | 102   | 92    | 79    | 66    | 63    |      |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 0.94   | 1.74  | 2.91  | 4.60  | 5.46  | 5.92  | 5.61  | 4.81  | 3.81  | 2.23  | 1.19  | 0.78  |      |
|                                                                        | RadStd      | 0.07   | 0.18  | 0.30  | 0.38  | 0.47  | 0.62  | 0.32  | 0.39  | 0.27  | 0.16  | 0.10  | 0.09  |      |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (26 neighbors) |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page